



# AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

**Alex**

FINAL SCORE

**16 / 70**

weighted points

CORRECT ANSWERS

**8 / 35**

questions

**NEEDS IMPROVEMENT**

Assessment Date: July 25, 2026      Report Generated: July 25, 2026 at 11:06 UTC      Role: Data Science Expert

OVERALL RATING

★☆☆☆☆

SUCCESS RATE

**22.9%**

WEIGHTED SCORE

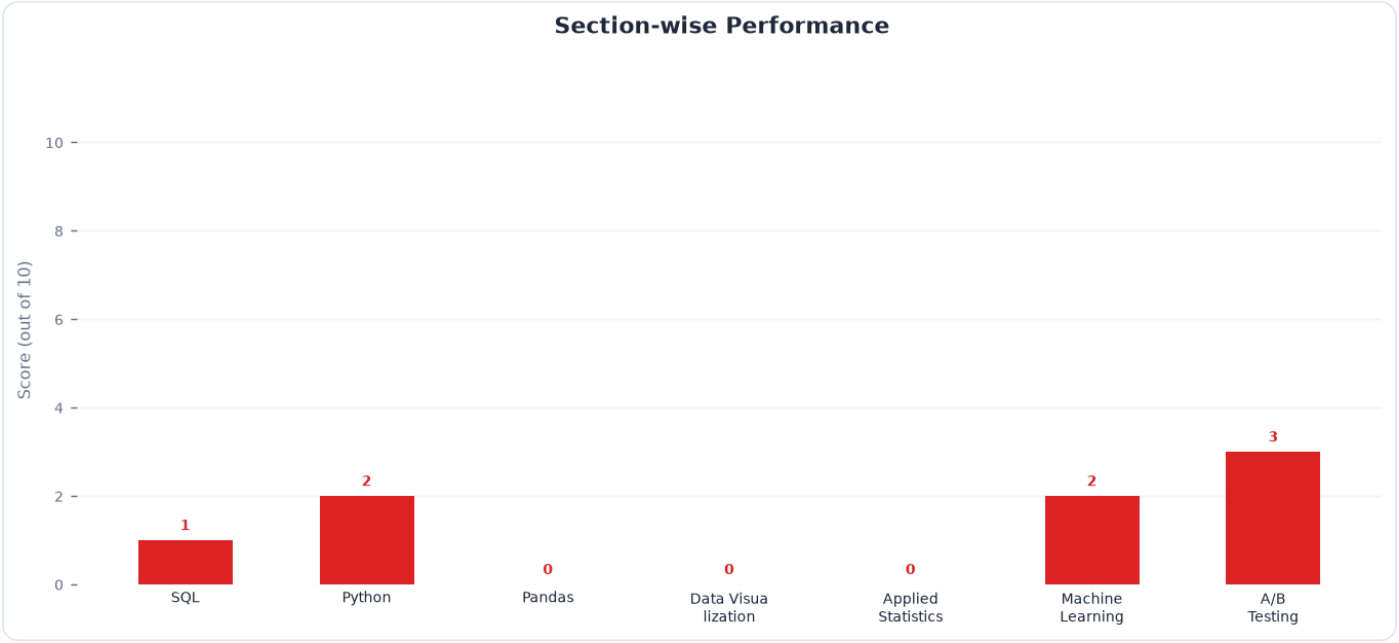
**16 / 70**

TOP SKILL

**A/B Testing**

NEEDS IMMEDIATE ATTENTION

**Pandas**



## Skill Rating & Benchmark

| Section            | Your Score | Reference Benchmark (Demo) | Delta | Status       |
|--------------------|------------|----------------------------|-------|--------------|
| SQL                | 1 / 10     | 6.8                        | -5.8  | Critical Gap |
| Python             | 2 / 10     | 6.2                        | -4.2  | Critical Gap |
| Pandas             | 0 / 10     | 5.8                        | -5.8  | Critical Gap |
| Data Visualization | 0 / 10     | 6.0                        | -6.0  | Critical Gap |
| Applied Statistics | 0 / 10     | 5.5                        | -5.5  | Critical Gap |
| Machine Learning   | 2 / 10     | 5.2                        | -3.2  | Needs Work   |
| A/B Testing        | 3 / 10     | 6.4                        | -3.4  | Needs Work   |

### Attempt Statistics Breakdown

| Section            | Total Q | Attempted | Correct | Incorrect | Skipped | Accuracy |
|--------------------|---------|-----------|---------|-----------|---------|----------|
| SQL                | 5       | 5         | 1       | 4         | 0       | 20%      |
| Python             | 5       | 5         | 2       | 3         | 0       | 40%      |
| Pandas             | 5       | 5         | 0       | 5         | 0       | 0%       |
| Data Visualization | 5       | 5         | 0       | 5         | 0       | 0%       |
| Applied Statistics | 5       | 5         | 0       | 5         | 0       | 0%       |
| Machine Learning   | 5       | 5         | 2       | 3         | 0       | 40%      |
| A/B Testing        | 5       | 5         | 3       | 2         | 0       | 60%      |

#### Executive Summary

The candidate achieved a score of 8 out of 35 correct answers (22.9% accuracy) with a weighted score of 16 out of 70. This performance indicates significant foundational gaps across SQL, Pandas, applied statistics, and machine learning. To secure a professional data science role, the candidate requires a structured upskilling plan focusing on core technical execution and statistical theory.

### Key Strengths

- Foundational Python Syntax:** Demonstrates familiarity with basic dictionary operations, string manipulation, and standard variable assignments.
- Basic Dataframe Slicing:** Understands the general purpose of Pandas indexing and column selection, despite minor errors with boundary inclusivity.

- **A/B Testing Terminology:** Shows awareness of core experimental concepts like statistical power, significance levels, and hypothesis testing structures.

Top Blind Spots

- **Advanced SQL & Window Functions:** The candidate struggles with complex query logic, including recursive CTEs, self-joins, and window functions like `ROW_NUMBER() OVER (PARTITION BY...)`.
- **Pandas Execution Details:** Gaps exist in understanding critical execution behaviors, such as label-based inclusive slicing in `.loc`, duplicate-dropping sorting logic, and specialized merges like `pd.merge_asof`.
- **Statistical Diagnostics & Hypothesis Testing:** Misunderstandings persist regarding Type I/II errors, ANOVA assumptions (homoscedasticity vs. normality), and the criteria for selecting Fisher's Exact Test over Chi-Square.

Actionable Learning Resources

**Complete the "Advanced SQL" track on Mode Analytics, focusing specifically on window functions, recursive CTEs, and self-joins.**

✓ Action Item

**Read Chapter 2 ("Pandas Essentials") of *Python for Data Analysis* by Wes McKinney to master indexing, duplicate handling, and time-series merging.**

✓ Action Item

**Study Chapter 3 ("Linear Regression") and Chapter 4 ("Classification") of *An Introduction to Statistical Learning (ISLR)* to reinforce hypothesis testing, ANOVA, and feature scaling.**

✓ Action Item

**Practice confusion matrix interpretation and model calibration techniques (like Isotonic Regression vs. Platt Scaling) using the Scikit-Learn documentation.**

✓ Action Item